

# PRG and PRF

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# Negligible Functions

PRG and PRF

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A function  $g(k)$  is negligible if  $\forall c > 0, \exists k'$  such that  $k \geq k'$

$$g(k) < \frac{1}{k^c}$$

- 1 let  $negl_1$  and  $negl_2$  are two negligible functions. Prove that  $negl_1 + negl_2$  is negligible.
- 2 Show that negligible function stays negligible even after polynomial attempts:  $poly(k)negl(k) = negl(k)$
- 3 If  $f(k)$  is not a negligible function and  $g(k)$  is a negligible function then show that  $f(k)g(k)$  is not a negligible function.

# Negligible Functions I

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**Claim.** Let  $\text{negl}_1$  and  $\text{negl}_2$  be two negligible functions. Then

$$\text{negl}_1(k) + \text{negl}_2(k)$$

is also negligible.

**Proof.** We need to prove that for every polynomial  $q(k)$ , there exists a value  $k'$  such that for all  $k \geq k'$ ,

$$\text{negl}_1(k) + \text{negl}_2(k) \leq \frac{1}{q(k)}. \quad (1)$$

Let  $q(k)$  be an arbitrary polynomial. Since  $\text{negl}_1$  is negligible, for the polynomial  $2q(k)$  there exists  $k_1$  such that for all  $k \geq k_1$ ,

$$\text{negl}_1(k) \leq \frac{1}{2q(k)}. \quad (2)$$

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Similarly, since  $\text{negl}_2$  is negligible, for the polynomial  $2q(k)$  there exists  $k_2$  such that for all  $k \geq k_2$ ,

$$\text{negl}_2(k) \leq \frac{1}{2q(k)}. \quad (3)$$

Let

$$k' = \max\{k_1, k_2\}.$$

Then, for every  $k \geq k'$ , inequalities (2) and (3) hold simultaneously. Therefore,

$$\text{negl}_1(k) + \text{negl}_2(k) \leq \frac{1}{2q(k)} + \frac{1}{2q(k)} = \frac{1}{q(k)}.$$

# Negligible Functions III

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Hence, for every polynomial  $q(k)$ , there exists  $k'$  such that for all  $k \geq k'$ ,

$$\text{negl}_1(k) + \text{negl}_2(k) \leq \frac{1}{q(k)}.$$

Thus,  $\text{negl}_1 + \text{negl}_2$  is a negligible function. □

**Claim.** Show that a negligible function stays negligible even after polynomial multiplication:

$$\text{poly}(k) \text{negl}(k) = \text{negl}(k).$$

**Solution.**

Let  $\text{negl}_1$  be a negligible function. Therefore,  
 $\forall c, \exists k'$  such that  $k \geq k'$

$$\text{negl}_1(k) \leq \frac{1}{k^c}.$$

Let  $k^{c_1}$  be an arbitrary polynomial. We need to prove

$$k^{c_1} \text{negl}_1(k) \leq \frac{1}{k^c}.$$

Since  $\text{negl}_1(k)$  is a negligible function, we have

$$\text{negl}_1(k) \leq \frac{1}{k^{c_1} \cdot k^c}.$$

Therefore,

$$k^{c_1} \text{negl}_1(k) \leq k^{c_1} \cdot \frac{1}{k^{c_1} \cdot k^c} = \frac{1}{k^c}.$$

Therefore,  $k^{c_1} \text{negl}_1(k)$  is a negligible function.

**Claim.** If  $f(k)$  is not a negligible function and  $g(k)$  is a negligible function, then

$$f(k) - g(k)$$

is not a negligible function.

**Proof.**

Assume,

$$f(k) - g(k)$$

is a negligible function. Let

$$f(k) - g(k) = \text{negl}(k).$$

Then,

$$f(k) = g(k) + \text{negl}(k).$$



Since  $g(k)$  is a negligible function and the sum of two negligible functions is negligible, it follows that

$$f(k)$$

is also a negligible function.

This contradicts the given assumption that  $f(k)$  is not a negligible function.

Therefore, our assumption is false. Hence,

$$f(k) - g(k)$$

is not a negligible function.



# Pseudorandom Generators I

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- 1** Let  $G_1, G_2 : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$  be two pseudorandom generators (PRGs) with  $G_1 \neq G_2$ . Define  $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$  by

$$G(s) = G_1(s) \oplus G_2(s),$$

Is  $G$  a PRG? Prove your answer.

- 2** Let  $G$  be a pseudorandom generator with expansion factor

$$\ell(n) > 2n.$$

In each of the following cases, say whether  $G_1$  is necessarily a pseudorandom generator. If yes, give a proof; if not, show a counterexample.

# Pseudorandom Generators II

## PRG and PRF

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**1** Define

$$G_1(s) = G(s_1 \cdots s_{\lfloor n/2 \rfloor}),$$

where

$$s = s_1 \cdots s_n.$$

**2** Define

$$G_1(s) = G(0^{|s|} \| s).$$

**3** Define

$$G_1(s) = G(s) \| G(s + 1).$$

**Claim.** Let  $G_1, G_2 : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$  be two pseudorandom generators (PRGs) with  $G_1 \neq G_2$ . Define  $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$  by

$$G(s) = G_1(s) \oplus G_2(s),$$

Is  $G$  a PRG? Prove your answer.

**Proof.** Given

$$G_1 : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$$

is a secure PRG. For any uniformly chosen seed

$$s \in \{0, 1\}^n,$$

define

$$G_2(s) = G_1(s) \oplus 1^{2n}.$$

We know that  $G_2$  is also a PRG. For the above definition of  $G_1$  and  $G_2$ , we construct a distinguisher  $D$  that distinguishes  $G$  from  $U_{2n}$ .

**Distinguisher  $D$ :**

**Input:**  $w \in \{0, 1\}^{2n}$

- If  $w = 1^{2n}$  then output 1.
- Otherwise output 0.

If

$$w \leftarrow G(s),$$

then

$$\Pr[D(w) = 1] = \Pr[D(G(s)) = 1] = 1.$$

If

$$w \leftarrow U_{2n},$$

then

$$\Pr[D(w) = 1] = \frac{1}{2^{2n}}.$$

Therefore,

$$|\Pr[D(G(s)) = 1] - \Pr[D(U_{2n}) = 1]| = \left| 1 - \frac{1}{2^{2n}} \right| > \text{negl}(n).$$

Hence,  $G$  is not a secure PRG.

(a)

$$G_1(s) = G(s_1 \cdots s_{\lfloor n/2 \rfloor}).$$

**Answer:**  $G_1$  is not a PRG as it does not satisfy the expansion condition.

We know that

$$G : \{0, 1\}^n \rightarrow \{0, 1\}^{\ell(n)}$$

is a PRG, where

$$\ell(n) > 2n.$$

Based on the construction of  $G_1$ , the domain is

$$\{0, 1\}^n$$

and the co-domain is

$$\{0, 1\}^{\ell(\lfloor n/2 \rfloor)}.$$

Assume

$$\ell(n) = 2n + 1.$$

Then

$$\ell(\lfloor n/2 \rfloor) = 2\lfloor n/2 \rfloor + 1 \leq 2\left(\frac{n}{2}\right) + 1 = n + 1.$$

Therefore,

$$\ell(\lfloor n/2 \rfloor) \leq n + 1.$$

For odd  $n$ ,

$$\ell(\lfloor n/2 \rfloor) = n.$$



Hence,  $G_1$  does not necessarily have output length greater than its input length and therefore does not satisfy the expansion requirement of a pseudorandom generator.

Thus,

$$G_1(s) = G(s_1 \cdots s_{\lfloor n/2 \rfloor}) \text{ is not a PRG.}$$

$$G_1(s) = G(0^{|s|} \| s)$$

First we define the PRG

$$G : \{0, 1\}^{2n} \rightarrow \{0, 1\}^{\ell(2n)},$$

such that

$$G(s_1) = \begin{cases} 0^{\ell(2n)}, & \text{if the first } n \text{ bits of } s_1 \text{ are zero,} \\ \hat{G}(s_1), & \text{otherwise, where } \hat{G} \text{ is a PRG.} \end{cases}$$

We can observe that  $G$  is a PRG.

If

$$w \leftarrow G(s_1),$$

the probability that a distinguisher distinguishes is

$$\frac{1}{2^n}.$$

Based on the above construction of  $G$ , we can construct the distinguisher  $D$ .

### Distinguisher $D$

Input:

$$w_1 \in \{0, 1\}^{\ell(2n)}$$

■ If

$$w_1 = 0^{\ell(2n)},$$

then output 1.

- Otherwise output 0.

If

$$w_1 \leftarrow G_1(s),$$

then

$$\Pr[D(w_1) = 1] = \Pr[D(G_1(s)) = 1] = \Pr[D(G(0^{|s|} \| s)) = 1] = 1.$$

If

$$w_1 \leftarrow U_{\ell(2n)},$$

then

$$\Pr[D(w_1) = 1] = \frac{1}{2^{\ell(2n)}}.$$

Therefore,

$$\left| \Pr[D(G_1(s)) = 1] - \Pr[D(U_{\ell(2n)}) = 1] \right| = \left| 1 - \frac{1}{2^{\ell(2n)}} \right|.$$

Hence,  $G_1$  is not a PRG.

# Pseudorandom Functions I

## PRG and PRF

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- 1 Consider the following keyed function  $F$ : For security parameter  $n$ , the key is an  $n \times n$  boolean matrix  $A$  and an  $n$ -bit boolean vector  $b$ . Define  $F_{A,b} : \{0,1\}^n \rightarrow \{0,1\}^n$  by  $F_{A,b}(x) = Ax + b \pmod{2}$ . Show that  $F$  is not a pseudorandom function.
- 2 Let  $F$  be a length-preserving pseudorandom function. For the following constructions of a keyed function

$$F' : \{0,1\}^{n-1} \rightarrow \{0,1\}^{2n},$$

state whether  $F'_k(x) = F_k(0\|x) \| F_k(1\|x)$  is a pseudorandom function.

Let  $O$  be an oracle that takes an  $n$ -bit string  $x$  as input and outputs either  $F_{A,b}(x)$  or  $F_n(x)$ , where  $F_n$  is a truly random function from  $\{0, 1\}^n \rightarrow \{0, 1\}^n$ .

Construct a distinguisher  $D$  that distinguishes  $F_{A,b}(x)$  from  $F_n(x)$ .

### Distinguisher $D$

- 1 Query the oracle  $O$  with inputs  $u$  and  $v$ , and compute

$$t = O(u) \oplus O(v) \oplus O(u \oplus v).$$

- 2 Similarly, query the oracle with inputs  $x$  and  $y$ , and compute

$$t_1 = O(x) \oplus O(y) \oplus O(x \oplus y).$$

3 If

$$t = t_1,$$

output 1. Otherwise output 0.

If the oracle outputs

$$w = F_{A,b}(x),$$

then

$$O(a) \oplus O(b) \oplus O(a \oplus b) = b$$

for all inputs  $a, b$ . Therefore,

$$t = t_1 = b,$$

and hence



$$\Pr[D(w) = 1] = 1.$$

If the oracle outputs

$$w = F_n(x),$$

where  $F_n$  is a truly random function, then  $t$  and  $t_1$  are independent uniformly random  $n$ -bit strings. Therefore,

$$\Pr[D(w) = 1] = \Pr[t = t_1] = \frac{1}{2^n}.$$

Since

$$|\Pr[D(F_{A,b}(x)) = 1] - \Pr[D(F_n(x)) = 1]| = \left|1 - \frac{1}{2^n}\right| > \text{negl}(n),$$

the distinguisher  $D$  distinguishes  $F_{A,b}$  from a truly random function with non-negligible advantage.

Therefore,  $F_{A,b}$  is not a pseudorandom function.



(a)

$$F'_k(x) = F_k(0\|x) \| F_k(1\|x).$$

**Claim:**  $F'_k$  is a pseudorandom function.

**Proof.**

Proof by Contradiction: Assume, that there exists a PPT distinguisher  $D$  that distinguishes

$$F'_k(x) = F_k(0\|x) \| F_k(1\|x)$$

from a truly random function

$$R : \{0, 1\}^{n-1} \rightarrow \{0, 1\}^{2n}$$

with non-negligible advantage.

We prove security using a hybrid argument.

Define the following hybrid distributions for a fixed input  $x$ :

$$H_0 = F_k(0 \| x) \| F_k(1 \| x),$$

$$H_1 = U_n \| F_k(1 \| x),$$

$$H_2 = U_n \| U_n,$$

where  $U_n$  denotes a uniformly random  $n$ -bit string.  
Observe that:

- $H_0$  is the output of  $F'_k(x)$ .
- $H_2$  is a uniformly random  $2n$ -bit string.

Suppose  $D$  distinguishes  $H_0$  from  $H_2$  with non-negligible advantage. By the hybrid argument, at least one of the following quantities must be non-negligible:

$$|\Pr[D(H_0) = 1] - \Pr[D(H_1) = 1]| ,$$

or

$$|\Pr[D(H_1) = 1] - \Pr[D(H_2) = 1]| .$$

### Case 1:

Suppose

$$|\Pr[D(H_0) = 1] - \Pr[D(H_1) = 1]|$$

is non-negligible.

Then we construct a distinguisher  $D_1$  against the PRF  $F$ . Given oracle access to a function  $O$ , where  $O$  is either  $F_k$  or a truly random function,  $D_1$  does the following:

- 1 Query  $O$  on  $0\|x$  and obtain  $y$ .
- 2 Independently compute  $F_k(1\|x)$ .
- 3 Give  $y\|F_k(1\|x)$  to  $D$ .
- 4 Output whatever  $D$  outputs.

If  $O = F_k$ , then the input given to  $D$  is distributed as  $H_0$ . If  $O$  is random, then the input given to  $D$  is distributed as  $H_1$ . Hence  $D_1$  distinguishes  $F_k$  from a random function with non-negligible advantage, contradicting the assumption that  $F$  is a PRF.

**Case 2:**

Suppose

$$|\Pr[D(H_1) = 1] - \Pr[D(H_2) = 1]|$$

is non-negligible.

Then we similarly construct a distinguisher  $D_2$  against  $F$ .  
Given oracle access to  $O$ :

- 1 Query  $O$  on  $1||x$  and obtain  $y$ .
- 2 Choose a uniform random string  $r \leftarrow U_n$ .
- 3 Give  $r||y$  to  $D$ .
- 4 Output whatever  $D$  outputs.

If  $O = F_k$ , then the input to  $D$  is distributed as  $H_1$ .

If  $O$  is random, then the input to  $D$  is distributed as  $H_2$ .

Thus  $D_2$  distinguishes  $F_k$  from a random function with non-negligible advantage, again contradicting the PRF security of  $F$ .

Since both cases lead to contradictions, neither hybrid transition can be distinguished with non-negligible advantage. Therefore,

$$H_0 \approx_c H_2.$$

Hence

$$F_k(0||x) || F_k(1||x)$$



is computationally indistinguishable from a uniformly random  $2n$ -bit string.

Therefore,

$$F'_k(x) = F_k(0\|x) \| F_k(1\|x)$$

is a pseudorandom function.

